

GIS Task

Prepare the Mandakini study-area vector layers and publish clean artifacts under `gis/outputs/` for backend and frontend consumption.

Task Specification: GIS, Hydrology & Geospatial Pipeline Lead

Role: GIS, Hydrology & Geospatial Data Engineer

Objective: Deliver clean, geographically accurate geospatial layers and terrain attributes for the **Mandakini River Basin (Rudraprayag District, Uttarakhand)** to power both the tactical frontend map and backend risk engine.

Milestone 1: College Demo Deliverables

Focus: Clean vector datasets that B (Frontend) and You (Backend) can immediately consume without waiting for heavy raster calculations.

- **Task 1.1: Study Bounding Box & Administrative Polygons**
- **Area of Interest (AOI):** Mandakini River Valley, centered around coordinates `30.28° N, 78.98° E`.
- Create `gis/outputs/villages.geojson` containing **3 distinct village/settlement polygons**:
- `VIL_TILWARA` (Tilwara)
- `VIL_SUMERPUR` (Sumerpur)
- `VIL_RUDRAPRAYAG` (Rudraprayag Town)
- **Required GeoJSON Property Schema:**

```
{
  "type": "Feature",
  "properties": {
    "village_id": "VIL_TILWARA",
    "name": "Tilwara",
    "population": 1840,
    "center_lat": 30.3522,
    "center_lon": 78.9634
  },
  "geometry": { "type": "Polygon", "coordinates": [...] }
}
```

- **Task 1.2: River Centerline & Safe Shelter Points**

- Extract/trace the Mandakini river path into `gis/outputs/river_mandakini.geojson` (LineString geometry).
- Mark 2–3 high-ground evacuation shelters in `gis/outputs/shelters.geojson` (Point geometries):
- `SHELTER_01`: "Government Inter College Grounds, Tilwara" (located well above the riverbed contour).
- `SHELTER_02`: "Rudraprayag Stadium Ground".
- **Task 1.3: Static Route Stub for Demo**
- Generate `gis/outputs/evac_route_tilwara.geojson` representing a clear, realistic road path from Tilwara village center uphill to `SHELTER_01`.
- *Action*: Push all GeoJSON files directly to `gis/outputs/` and notify Frontend builder so they can render them on Leaflet/Mapbox.

Milestone 2: Post-Demo System Expansion

Focus: Deep topographic physics, DEM processing, and dynamic road graph networks for the final SIH submission.

- **Task 2.1: DEM Acquisition & Topographic Indices**
- Download the **Copernicus 30m GLO-30 DEM** GeoTIFF covering the Rudraprayag catchment tile from OpenTopography or USGS EarthExplorer.
- Store raw rasters under `gis/raw/dem_rudraprayag_30m.tif`.
- Write a Python pipeline using `rasterio` and `richdem` (`gis/src/process_terrain.py`) to compute:
 1. **Slope Gradient Matrix (β)**: Identifies steep runoff inclines.
 2. **Flow Direction & Flow Accumulation Matrix (a)**: Identifies natural drainage pathways.
 3. **Topographic Wetness Index (TWI)**:

$$TWI = \ln \left(\frac{a}{\tan \beta} \right)$$

- Export a normalized summary table `gis/outputs/village_terrain_features.csv` with mean slope and average TWI per village polygon to supply features to the backend ML model.
- **Task 2.2: Road Network Graph Extraction**
- Use `osmnx` to download the drivable/walkable road network for the bounding box:

```
import osmnx as ox
G = ox.graph_from_bbox(north=30.42, south=30.25, east=79.05, west=78.92,
network_type="all")
ox.save_graphml(G, "gis/outputs/rudraprayag_roads.graphml")
```

- Map which road segments fall within the high-TWI or flood-prone buffer zone (within 50 meters of the river centerline).

- Deliver `rudraprayag_roads.graphml` to the backend lead for risk-weighted Dijkstra edge cost calculations.
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Verification & Quality Checklist

- ☐ All coordinates must use standard **WGS84 (EPSG:4326)** projection.
- ☐ GeoJSON files validated on geojson.io before committing.
- ☐ File sizes for vector files must be compact (< 2 MB) for smooth frontend rendering.
- ☐ Python scripts must include a `requirements.txt` (`geopandas`, `rasterio`, `shapely`, `osmnx`, `richdem`).